LEARNING WHAT TO CHANGE

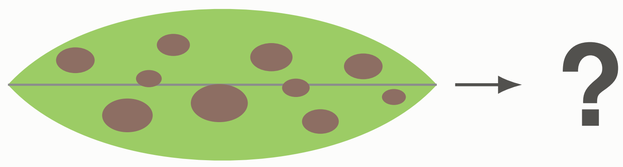
From Scientific Observations to Executable Reasoning and Hypothesis Revision

Chimdi Walter Ndubuisi · University of Missouri REVIEW DRAFT 19 October: for collaborator review, not for print

How can a machine learn the objects, relationships and rules in unfamiliar data—and know what to reconsider when its explanation is not enough?

Biology illustrates the difficulty: measurements do not arrive with the correct entities, mechanisms or reasoning rules already identified.

ILLUSTRATIVE SCHEMATIC, not data

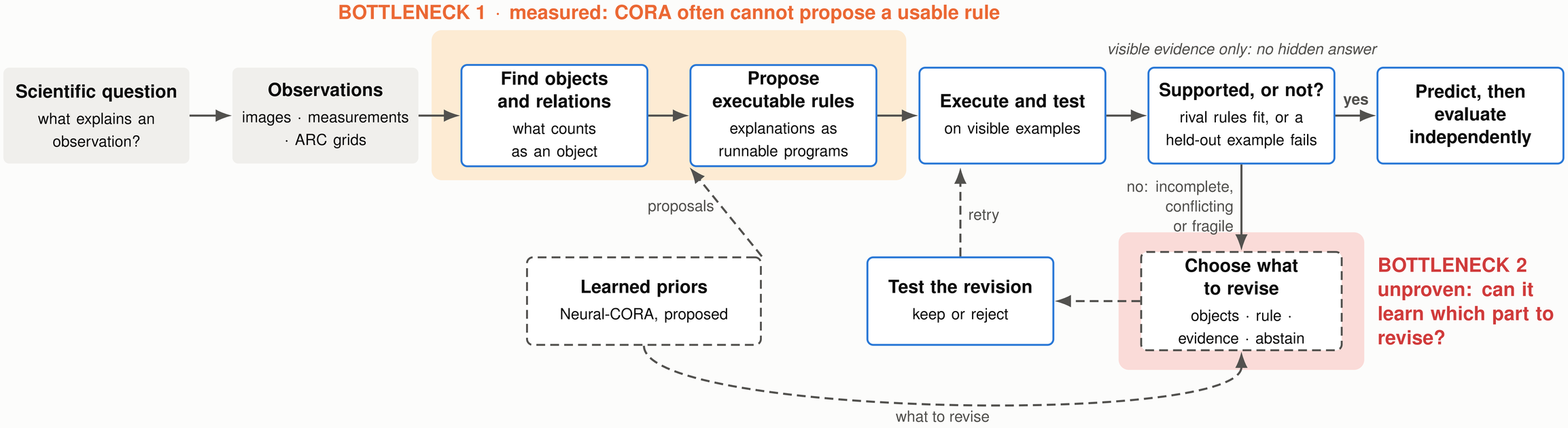


maize lesion-mimic pattern

What process could explain these patterns?

Observed pattern ≠ identified mechanism.

One reasoning loop

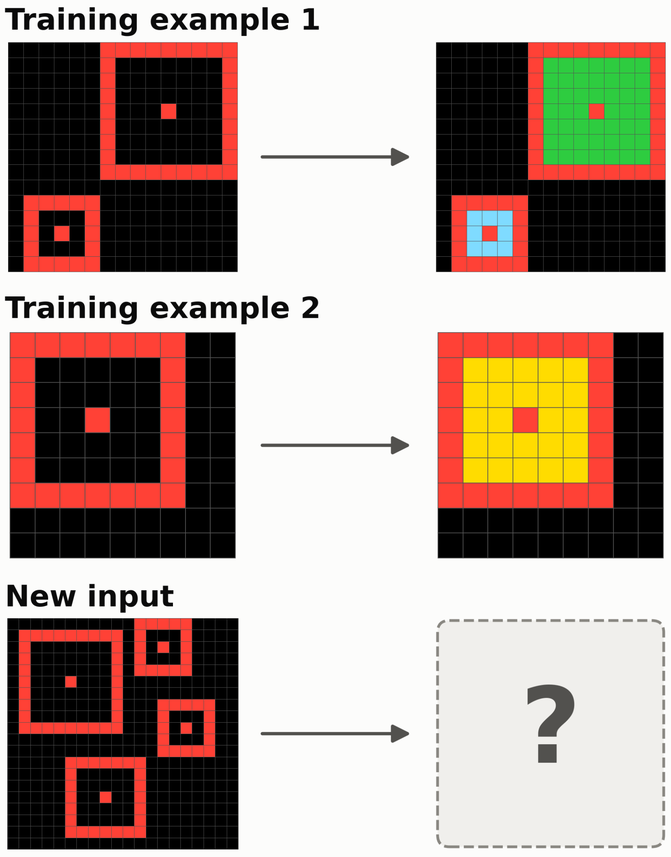


Solid: implemented and tested. Dashed: proposed. Grey: context. A tested revision is not automatically a correct explanation.

ARC is our controlled testbed

Few examples, a new input, one exact answer.

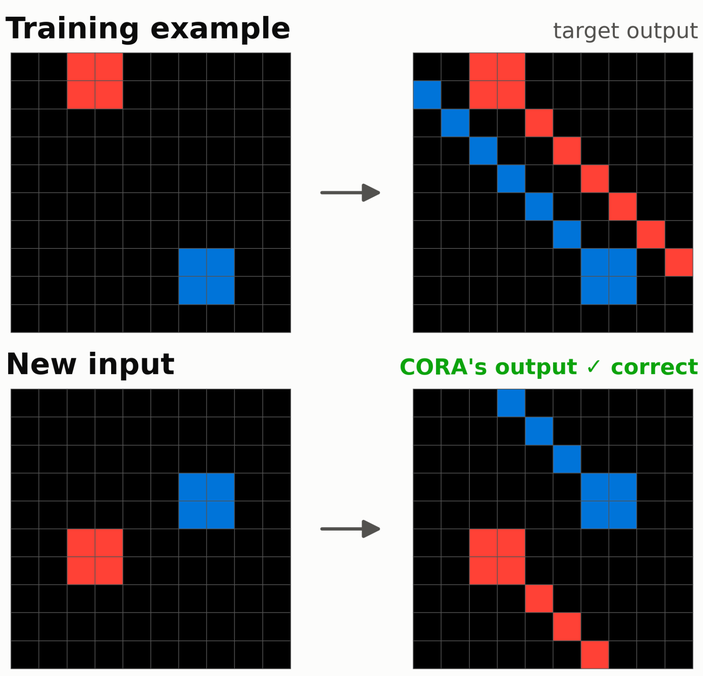
REAL ARC-AGI-2 TRAINING TASK 00dbd492 (public)



Rule: each frame's fill colour depends on its size.

Done: a rule CORA built [DONE, NARROW]

REAL ARC-AGI-2 TRAINING TASK ka95aa62f67d9 · recorded run

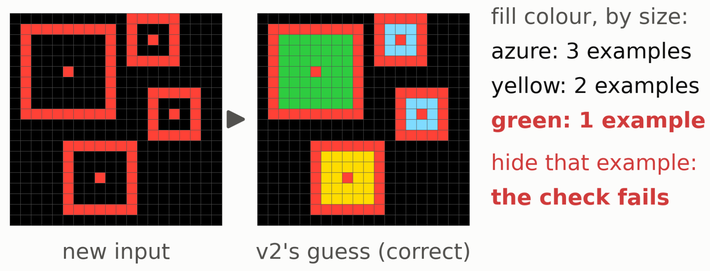


Rule: **blue** objects send a line up-left; **red** ones, down-right.

Rebuilt 3 of 3 times with one example hidden; correct on an exposed training task, not a competition solve. Also found without the failure record.

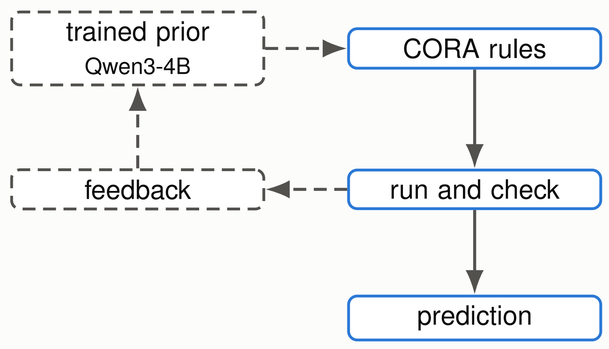
Visible doubt [DEVELOPMENT]

REAL TASK 00dbd492 · CORA-v2 development run
CORRECT DEVELOPMENT PREDICTION — NOT CERTIFIED



One example decides the green fill, so CORA-v2 keeps its correct guess uncertified.

Next: learning inside CORA [PROPOSED]

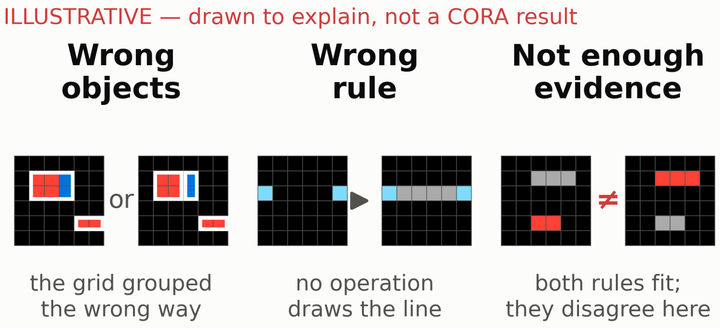


A trained prior would only order CORA's own search. No result yet.

NVARC (published Qwen3-4B method), reproduced: 27.64% public. Native CORA and CORA with rule construction: 0.00% public.

Why rules fail

ILLUSTRATIVE — drawn to explain, not a CORA result



Memorization: fits the examples, fails the new input.

Where it fails [NEGATIVE]

0 / 58

new verified solutions from rule construction

REAL · 58 separately selected, exposed training tasks



Every variant: 0; no wrong answers.

If revision selection works on ARC, test the reasoning procedure—not ARC grid operations—on independently validated biological questions. [UNTESTED]

Prior work: Popper (Cropper & Morel 2021), DreamCoder (Ellis et al. 2021), LILO (Grand et al. 2024), Model Discovery Agent (Murphy 2026); details in the explainer. Numbers: `records/CORA_SUMMIT_CLAIM_LEDGER_20261017.md`. AI coding assistance used. Authorship pending collaborator review.